Graphing Basic Polar Curves from Equations

Evaluating r at a given angle, finding the angles where r is zero, locating the maximum r that fixes a curve's orientation, and identifying and sketching basic polar curves.

- Work in degrees. Where an interval is asked for, use $0° \leq \theta < 360°$ and list every solution in that window.
- Give exact values, not decimals.
- A negative r is legal: plot $|r|$ in the direction opposite θ.
- Show the equation you solved, not just the angles.

1. For the circle $r = 6\cos\theta$, find r when $\theta = 60°$.

Answer: ________________

2. For the circle $r = 4\sin\theta$, find r when $\theta = 30°$.

Answer: ________________

3. For the cardioid $r = 1 + \cos\theta$, find r when $\theta = 90^\circ$. This value is the curve's distance from the pole straight up the 90° ray.

Answer: ______________

4. The rose $r = 4\sin(2\theta)$ returns to the pole between petals.
Find every θ in $0^\circ \leq \theta < 360^\circ$ for which $r = 0$. These angles are the lines the petals are tucked between.

Answer: ______________

5. For the limacon $r = 2 + 4\cos\theta$, find r when $\theta = 180^\circ$. Read your answer carefully: the sign matters.

Answer: ______________

6. For the cardioid $r = 1 + \cos\theta$, find every θ in $0° \leq \theta < 360°$ with $r = 0$. This single angle is where the cardioid's cusp sits.

Answer: ________________

7. The rose $r = 2\sin(3\theta)$ reaches its greatest distance from the pole when $r = 2$. Solve $2\sin(3\theta) = 2$ for every θ in $0° \leq \theta < 360°$. These angles point along the petal tips.

Answer: ________________

8. For the limacon $r = 3 - 2\cos\theta$, find r when $\theta = 120°$.

Answer: ________________

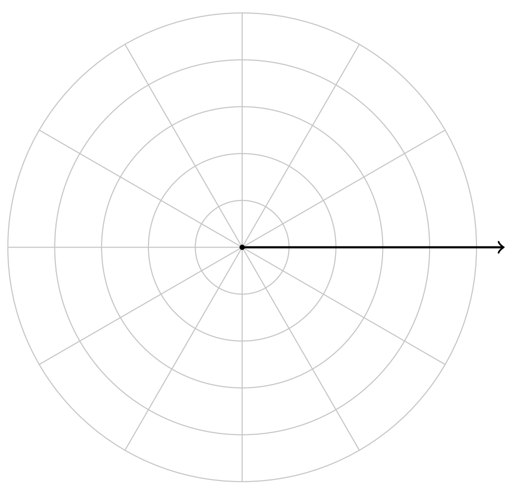
9. The limacon $r = 2 + 4\cos\theta$ passes through the pole twice, and those two angles bound its inner loop.
Find every θ in $0° \le \theta < 360°$ for which $r = 0$.

Answer: ________________

10. For the circle $r = 4\cos\theta$, the maximum value of r is 4.
Solve $4\cos\theta = 4$ on $0° \le \theta < 360°$. The single angle you find is the direction in which the circle is offset from the pole.

Answer: ________________

11. **Sketch.** Graph the cardioid $r = 1 + \cos\theta$ on the polar grid below. Plot the four points at $\theta = 0°$, $90°$, $180°$, and $270°$ first, then join them smoothly. Label the cusp.
Each ring is 1 unit; the outermost ring is $r = 5$. The heavy ray is the polar axis, $\theta = 0°$.



12. Identify, sketch, and justify. Consider $r = 2 + 4\cos\theta$.
Name the family this curve belongs to, sketch it on the grid below, and explain — using the constant term and the coefficient of $\cos\theta$ — why this curve has an inner loop while $r = 4 + 2\cos\theta$ would not. Use the pole-crossing angles you found earlier as evidence.
Each ring is 1 unit; the outermost ring is $r = 6$. The heavy ray is the polar axis, $\theta = 0°$.